

ABSTRACT

This project represents the practical application of E-Commerce principles, rooted in the foundational concepts of the MIS & E-Business course from the fifth semester, with a focus on B2C commerce strategies. The initiative aimed to design and develop an online platform that delivers an engaging shopping experience while addressing the growing demand for digital marketplaces. By implementing features such as product categorization, shopping cart functionality, and secure payment processing, the project exemplifies the integration of technology and business objectives. This endeavor has not only deepened our understanding of E-Commerce operations but also reinforced the importance of user-centric design, innovation, and collaboration in creating impactful digital solutions for modern consumers.

Keywords: E-Commerce, B2C commerce, MIS, E-Business, digital marketplace, user-centric design, secure payment, online shopping.

Acknowledgment

We would like to express our sincere gratitude and appreciation to all those who have contributed to the completion of this project.

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List of Abbreviations

B: Business

C: Consumer

G: Government

IRD: Inland Revenue Department

SEO: Search Engine Optimization

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Chapter 1: INTRODUCTION

1.1 Introduction to E-Commerce

E-commerce is the process of purchasing and selling products or services over the Internet. It enables companies to establish virtual stores on websites or apps, allowing customers to purchase whenever and wherever they want. Payments are done electronically, and goods are shipped to homes. This is a modern and convenient way to do business without having a physical store.



Figure 1. 1 : E-commerce cycle

1.2 Types of E-Commerce

1. Business-to-Consumer (B2C)
2. Business-to-Business (B2B)

3. Consumer-to-Consumer (C2C)
4. Consumer-to-Business (C2B)
5. Government-to-Business (G2B)
6. Government-to-Consumer (G2C)

1.2.1 Business-to-Consumer (B2C)

In e-commerce, businesses sell their products or services directly to the end-buyer, the consumer. It prioritizes convenience, permitting people to shop anytime and anywhere. Companies display their goods online, and buyers are able to order and have things delivered to their homes.

For example: Buying things like clothes or electronics in Nepal from Daraz.

1.2.2 Business-to-Business (B2B)

B2B refers to businesses that sell when other businesses. These tend to be large-scale transactions that allow companies to acquire the tools, raw materials, or services necessary for operations. It is relationship-driven with long-term partnerships and hence, repeat orders.

For example: A Nepali wholesale supplier supplying raw materials to food production companies.

1.2.3 Consumer-to-Consumer (C2C)

C2C e-commerce allows you to sell your products directly to the buyer, Consumer-to-Consumer. Those services are easy to use, so people often use online platforms to sell used goods, homemade products, or services. It's a frugal way to trade and save.

For example: Selling a second-hand motorbike on HamroBazar.

1.2.4 Consumer-to-Business (C2B)

As for the "C2B", it stands for "Customer to Business" which means you are offering your skills or services to a business. Individuals use platforms to sell their expertise to businesses that need skills such as graphic design, writing or photography. This enables people to work freelance and get rewarded for their ideas.

For example: A Nepali freelancer creating design banners for promotional use for local businesses.

1.2.5 Government-to-Business (G2B)

Governments purchase goods or services from businesses to fulfill their needs. These contracts can cover the building of infrastructure, the supply of office equipment, or the establishment of IT systems. These are services that businesses provide under formal contracts to assist the government in its operations.

For example: Nepali government buys software of e-governance systems from local IT company.

1.2.6 Government-to-Consumer (G2C)

Citizens perform transactions with the government online paying taxes, renewing licenses, or applying for permits. Unlike other transactions, this process is quick and saves time. State-run online portals prove efficient and transparent, allowing citizens to undertake essential tasks from home.

For example, paying taxes or renewing a driver's license through Nepal's IRD portal.

1.3 Objectives

- To sell products online through the ecommerce system,
- To integrate secure payment gateways,
- To manage product inventory effectively,

Chapter 2: Process of Building a Website

2.1 Steps for Building a Website

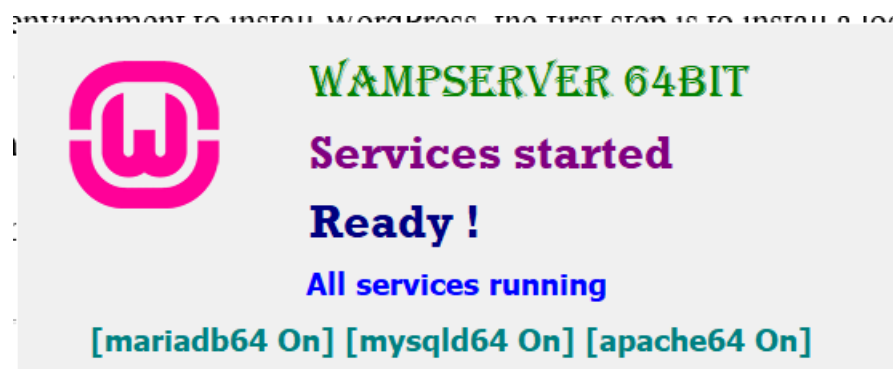
2.1.1 Choose a Domain Name and Hosting

A domain name that reflects the website's purpose is selected. A hosting provider is chosen based on the needs and budget. One-click WordPress installation is often provided by many hosting providers to simplify the setup process.

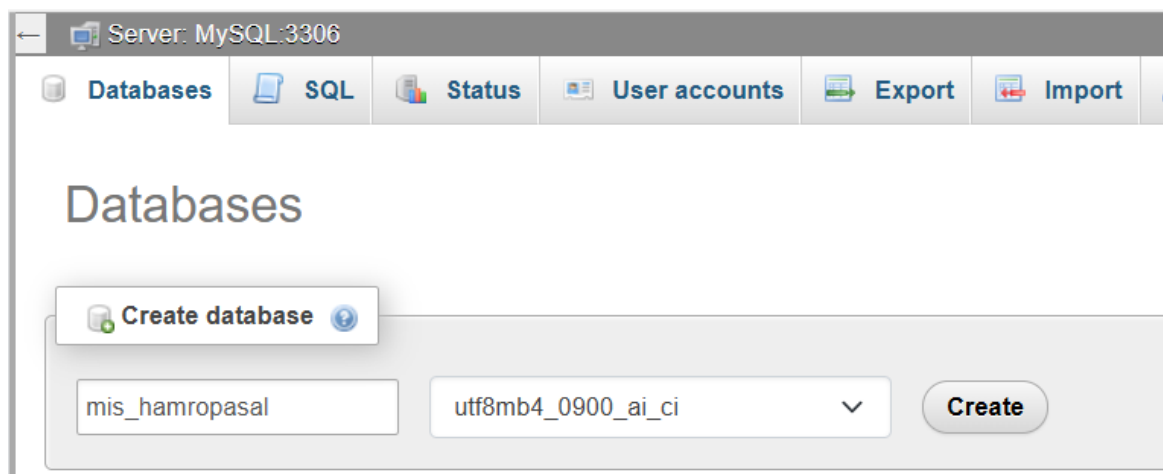
2.1.2 Install WordPress

Once the domain and hosting are secured, WordPress is installed on the server. Easy installation options, such as Softaculous, are provided by most hosting services. In the Local environment to install WordPress, the first step is to install a local server stack like XAMPP, WAMP, MAMP, or Local by Flywheel, which includes PHP, MySQL, and Apache/Nginx.

- **Download & Install WAMPP**

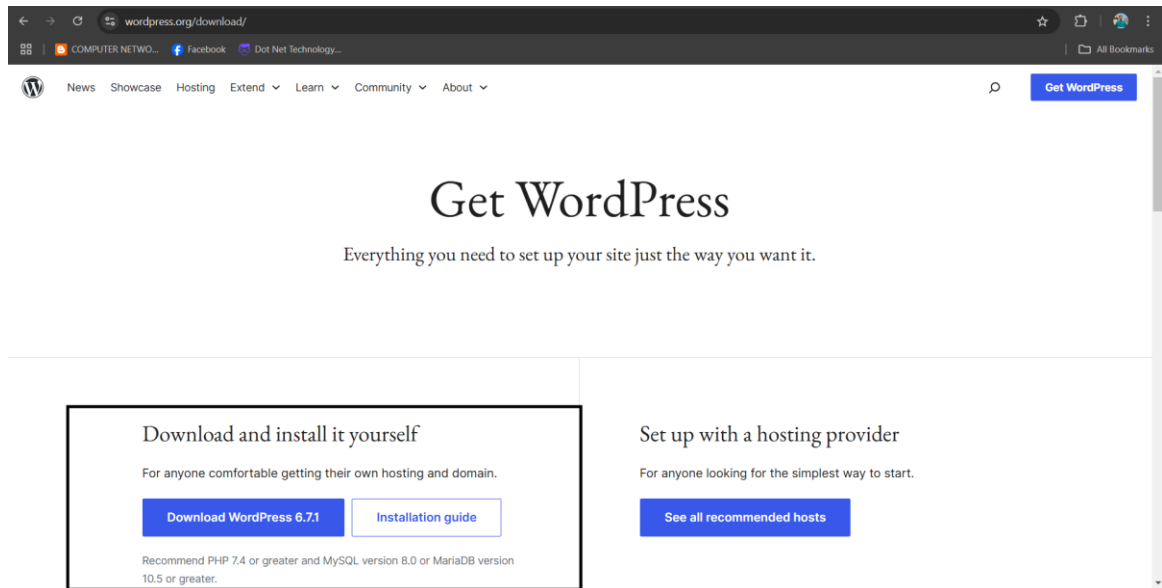


- **Start the Local Server**
- **Create a Database**



Open phpMyAdmin (<http://localhost/phpmyadmin/>), click on new and enter name of database and click on create button.

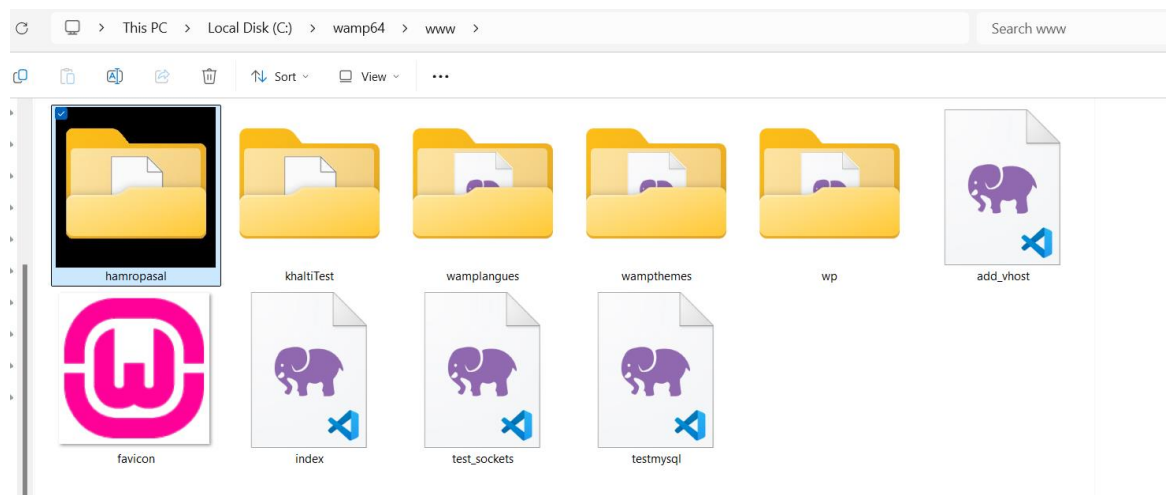
- **Download WordPress**



Goto <https://wordpress.org/download/> and click on **Download Wordpress** button. Then a zip file will be downloaded. [1]

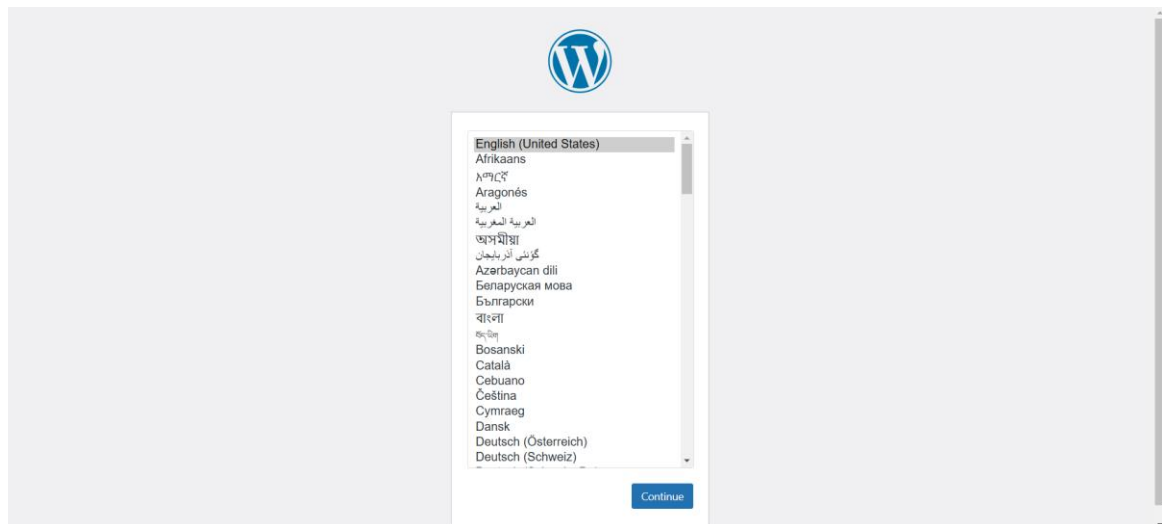
- **Move WordPress Files to the Local Server**

Once extracted, copy the WordPress folder and paste it into the **www** directory within the WAMPP installation folder:



- **Configure WordPress**

Open <http://localhost/wordpress> in the browser.



Choose the desired language and click on continue,

Below you should enter your database connection details. If you are not sure about these, contact your host.

Database Name
The name of the database you want to use with WordPress.

Username
Your database username.

Password [Hide](#)
Your database password.

Database Host
You should be able to get this info from your web host, if localhost does not work.

Table Prefix
If you want to run multiple WordPress installations in a single database, change this.

[Submit](#)

Click on Submit and run the installation. Complete the installation by entering **Site Title**, **Username**, **Password**, **Email** then click Install WordPress. After installation is complete, visit <http://localhost/⟨⟨folder-name⟩⟩/wp-admin> and log in using the created credentials.

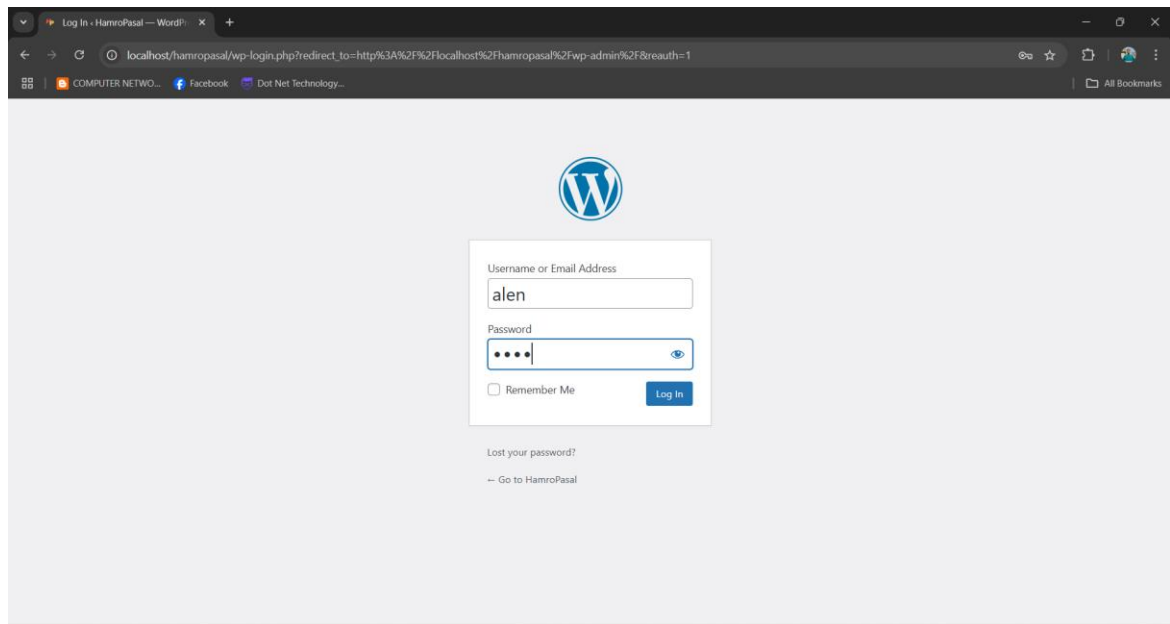


Figure 2.1 Install WordPress

2.1.3 Select a Theme

A WordPress theme that aligns with the website's design and purpose is selected. Free and premium themes can be found in the WordPress Theme Directory or third-party marketplaces. The theme is installed and activated through the WordPress dashboard. [2]

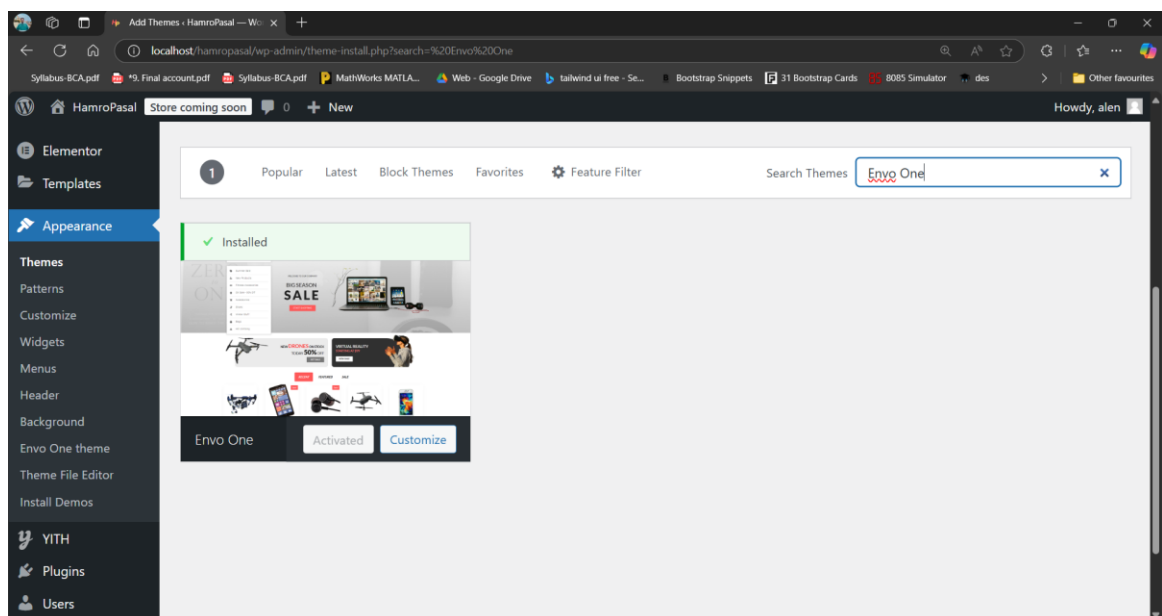


Figure 2.2 Select a Theme

Envoy One is a fast, clean, lightweight, customizable and modern-looking free responsive WooCommerce WordPress theme for e-commerce stores like clothing, fashion, electronics,

ebooks, dress store, sports shop, marketplace, multivendor, equipments or online products. Theme works perfectly with the new block editor (Gutenberg) and popular page builders like Elementor, Beaver Builder, SiteOrigin, Thrive Architect, Divi, Brizy, Visual Composer etc. Theme is compatible with YITH WooCommerce Wishlist, YITH WooCommerce Compare, WooCommerce Variation Swatches and other popular plugins.

2.1.4 Customize your Website

The website's appearance is customized using the WordPress Customizer and theme options. Colors, fonts, and layouts are adjusted to match the desired design. Branding elements like the logo are added to give the website a professional look.

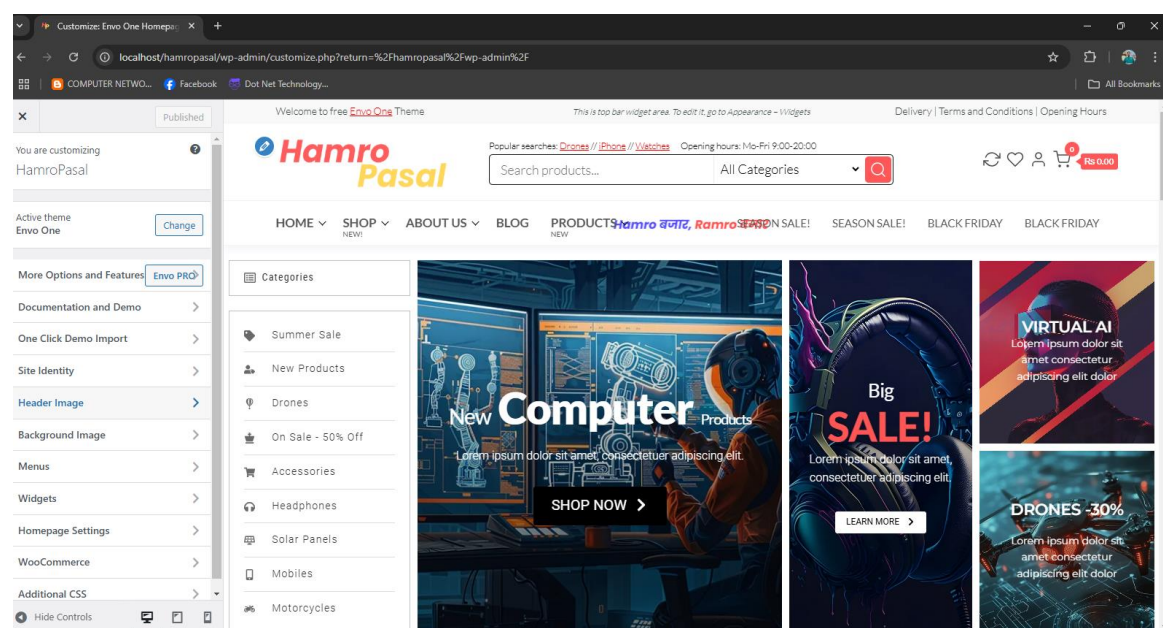


Figure 2.3: Customize your Website

To customize the Envo One theme in WordPress, log in to your dashboard, navigate to **Appearance > Customize**, and use the WordPress Customizer to make changes with a live preview. You can adjust the **Site Identity** (title, tagline, and logo), modify colors, typography, and background settings, and configure WooCommerce options like product displays and checkout layouts for online stores. Customize the header and footer by adding menus, widgets, or images, and refine sidebars or footer areas using **Appearance > Widgets**. For advanced layouts, use a page builder like Elementor or Divi to add product grids, sliders, or custom elements. Enhance functionality with compatible plugins such as YITH WooCommerce Wishlist, WooCommerce Variation Swatches, and SEO tools. Once your changes are complete, click Publish to save and apply them to your site.

2.1.5 Add Content

Content is added to the website, including essential pages such as Home, About, and Contact. The WordPress editor is used to format text, add images, and organize content effectively for users.

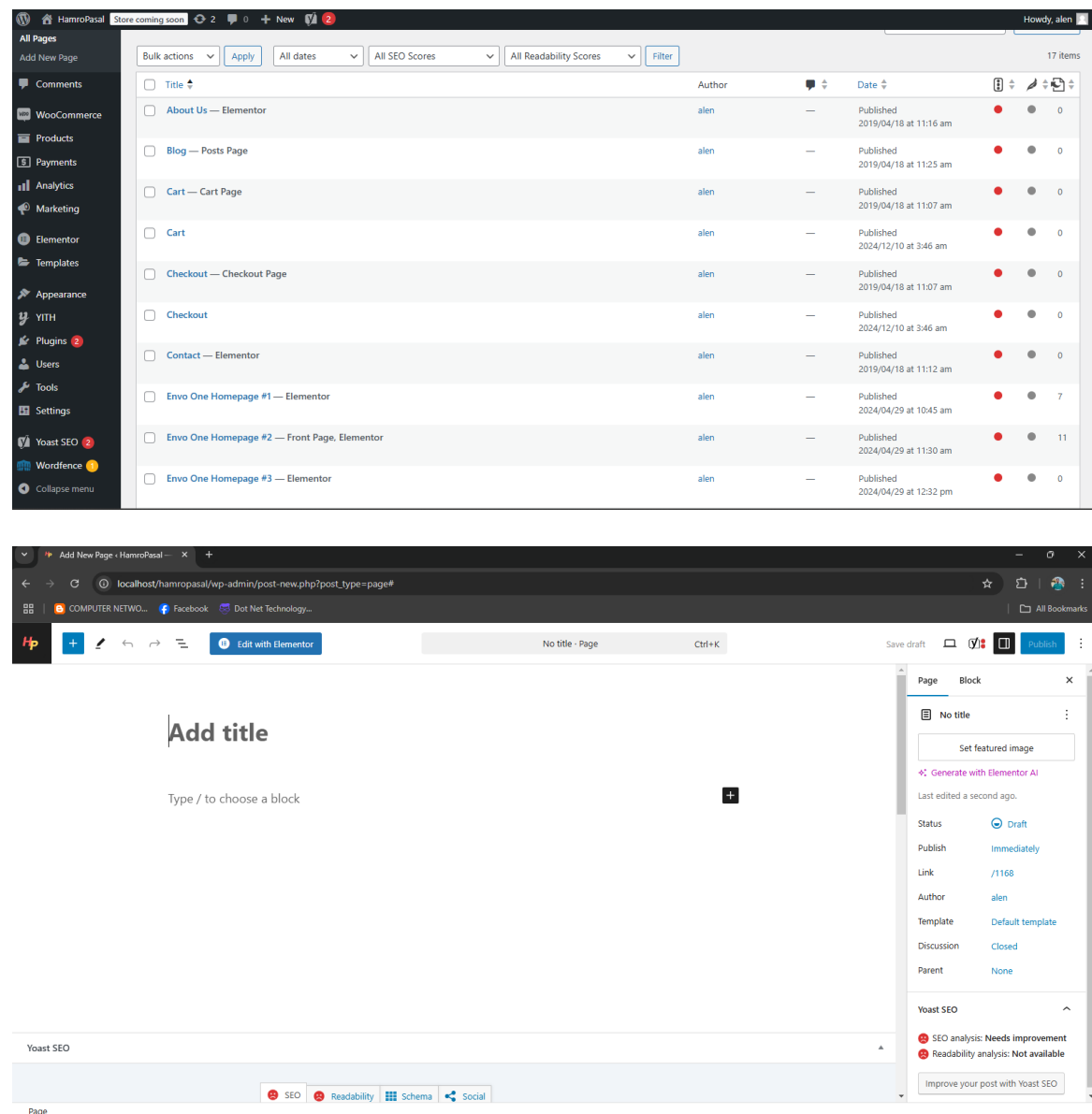


Figure 2.4: Add Content

To add content as a page in WordPress, go to your dashboard, navigate to **Pages > Add New**, and enter a title for the page. Use the block editor (Gutenberg) to add text, images, videos, and other elements by clicking the + icon to insert blocks. Customize the layout by arranging blocks and configuring their settings on the right sidebar. Add a featured image if needed, set the page attributes like parent page or order, and adjust visibility or publication status. Once done, click Publish to make the page live on your site.

2.1.6 Install Plugins

To enhance the functionality of your website, plugins are installed. Plugins are tools that add extra features to your site, such as improving SEO, adding contact forms, or enabling e-commerce. For example, you can install WooCommerce to sell products online, Yoast SEO for better search engine visibility, or Elementor to build website components. Plugins are easily installed through the WordPress dashboard, where you can search for, install, and activate them to extend your website's capabilities.

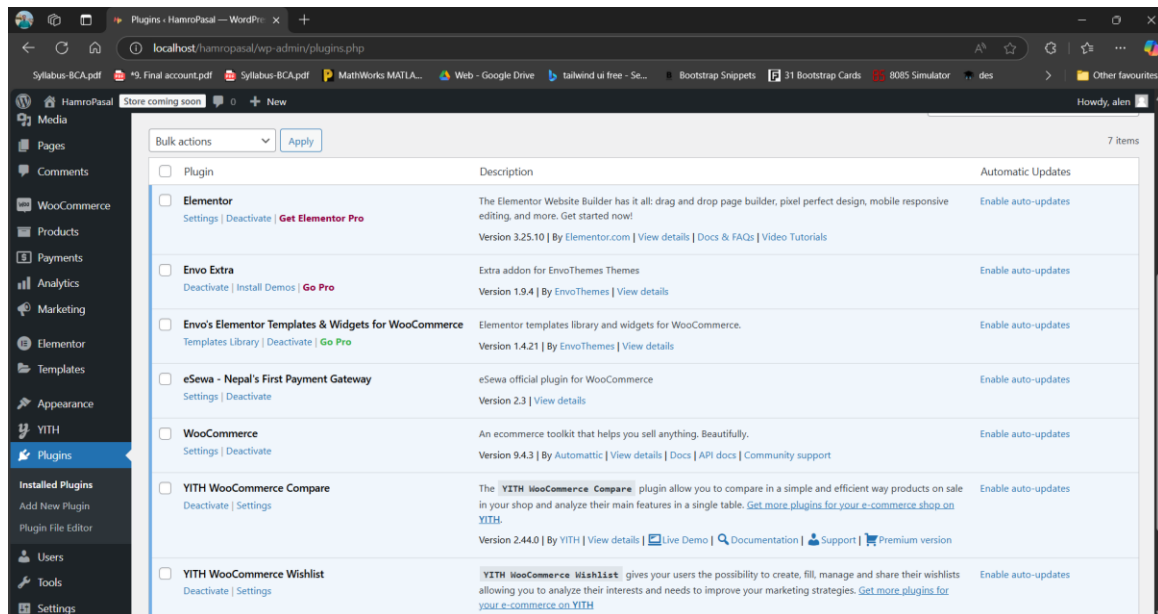


Figure 2.5: Install Plugins

To install a plugin in WordPress, log in to your dashboard and navigate to **Plugins > Add New**. Use the search bar to find the desired plugin or upload a plugin file by clicking **Upload Plugin** and selecting a .zip file. Once the plugin appears, click **Install Now**, wait for the installation to complete, and then click **Activate** to enable it on your site. The plugin is now ready to use, and you can configure its settings as needed.

2.1.7 Optimize for SEO

SEO optimization is done to improve the website's visibility in search engine results. An SEO plugin, like Yoast SEO, is used to optimize page titles, meta descriptions, and keywords. A sitemap is created to assist search engines.

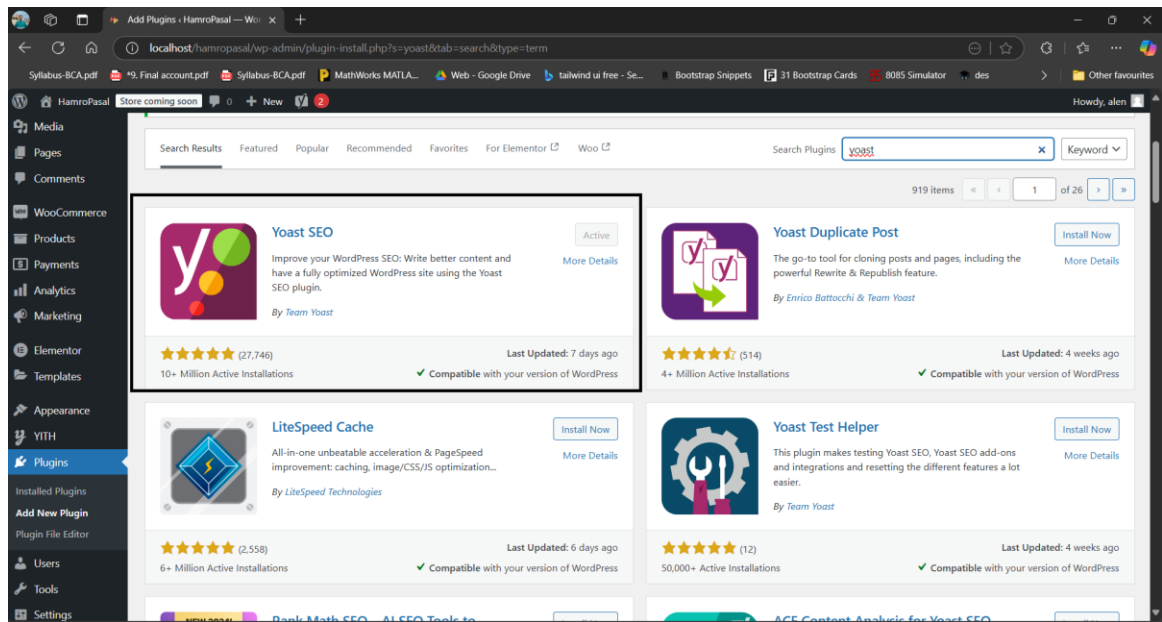
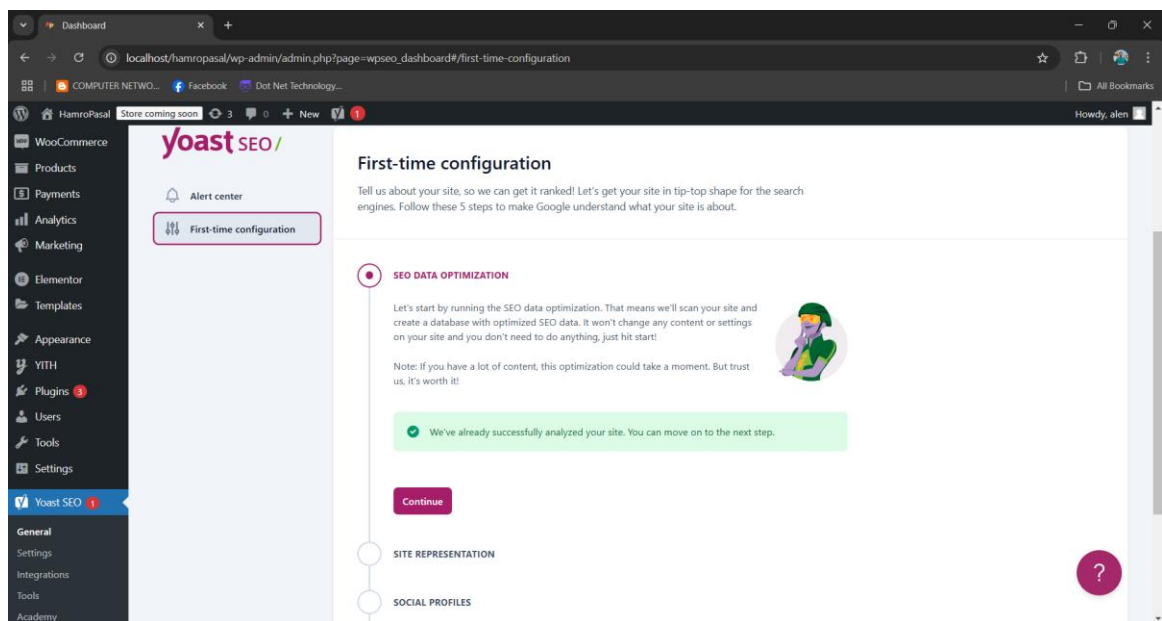


Figure 2.6: Optimize for SEO



To optimize a WordPress site using Yoast SEO, the user should first install and activate the plugin through **Plugins > Add New**. Basic settings can be configured using the setup wizard under **SEO > General**, where the site type can be specified, and features like XML sitemaps can be enabled. When editing posts or pages, the Yoast SEO meta box allows the user to enter a focus keyphrase, optimize the SEO title and meta description, and follow content analysis suggestions to improve keyword usage, readability, and text structure. Yoast automatically generates an XML sitemap, which can be submitted to Google Search Console for better search engine indexing. Additionally, the user can customize social media previews under the **Social** tab to enhance how content appears when shared. With

these steps, Yoast SEO simplifies and enhances on-page optimization. Key features include on-page SEO optimization with real-time content analysis and keyword suggestions, automatic XML sitemap generation, customizable title and meta descriptions, and social media integration for better sharing. It also offers readability analysis, breadcrumb navigation, internal linking suggestions, and permalink controls for a cleaner URL structure.

2.1.8 Set up Security

Security measures are implemented to protect the website. Strong passwords are used, and WordPress, themes, and plugins are kept up to date. A security plugin like Wordfence is installed, and regular backups are scheduled to prevent data loss.

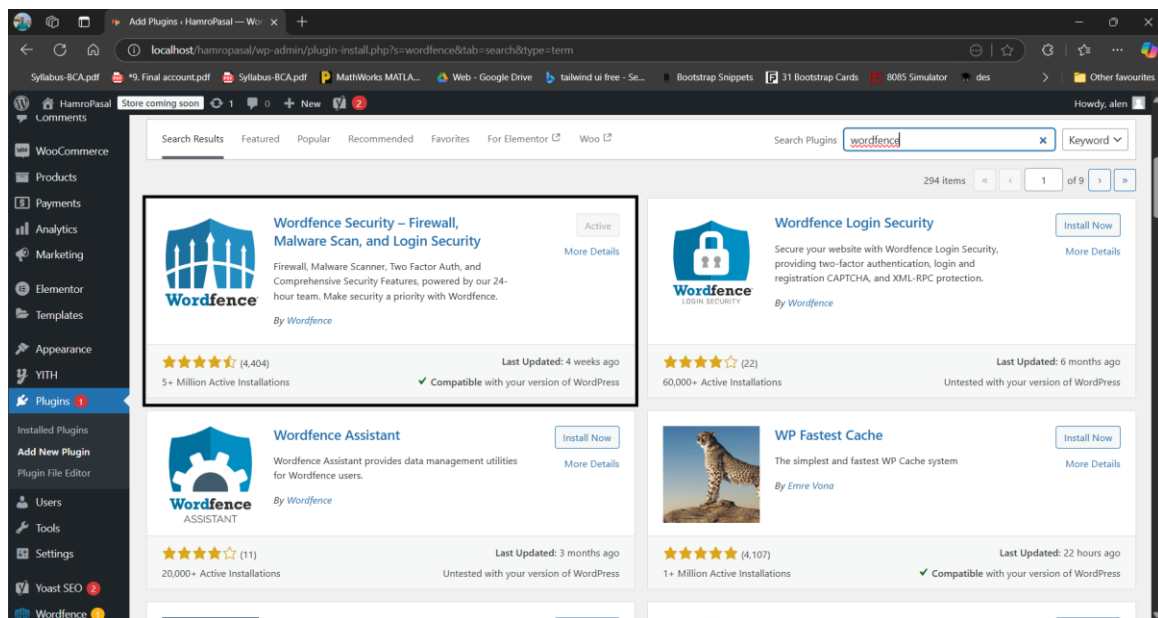


Figure 2.7: Set up Security

To enhance WordPress security, the user can install the **Wordfence Security** plugin by going to **Plugins > Add New**, searching for Wordfence, and activating it. Once installed, Wordfence provides real-time threat defense, firewall protection, and malware scanning. It also offers login security features like two-factor authentication and limits login attempts to prevent brute-force attacks. The user can configure advanced settings for blocking malicious IPs, monitoring live traffic, and conducting regular scans for vulnerabilities. Additionally, Wordfence provides detailed security reports and alerts, helping the user stay informed and secure their website from potential threats.

2.1.9 Test and Launch

The website is thoroughly tested before launching. Broken links are checked, and the website's responsiveness across different devices is reviewed. Any issues are fixed before launching the website.

2.1.10 Maintain and Update

Regular maintenance is carried out by updating WordPress, themes, and plugins. Website performance is monitored, and issues are addressed promptly to ensure a smooth user experience. New content is added periodically to keep the website fresh and engaging.

2.2 Procedures during Design of Ecommerce Store

2.2.1 Adding Products

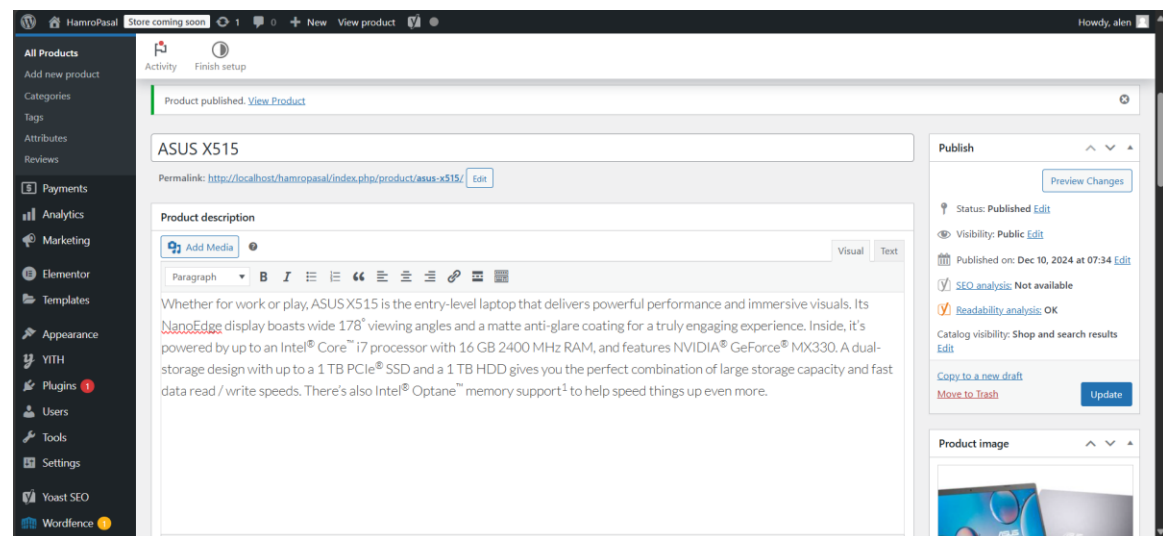


Figure 2.8: Adding Products

To add a product in WooCommerce on WordPress, the user should go to **Products > Add New** in the WordPress dashboard. Here, they can enter the product title and description. Under the **Product Data** section, the user can select the product type (simple, variable, grouped, or external/affiliate). For simple products, the user can enter the regular price, sale price, and SKU (if necessary). Additionally, users can manage stock, set shipping details, and add product attributes or variations for variable products. In the **Product Short Description** section, the user can add a brief summary that will appear on the product page.

The user can also assign the product to a category (or create a new one), add tags, and upload product images and galleries. Once all the necessary details are filled out, the user

can click the **Publish** button to make the product live on the store. The product can then be viewed on the website, and customers can purchase it.

2.2.2 Adding Categories

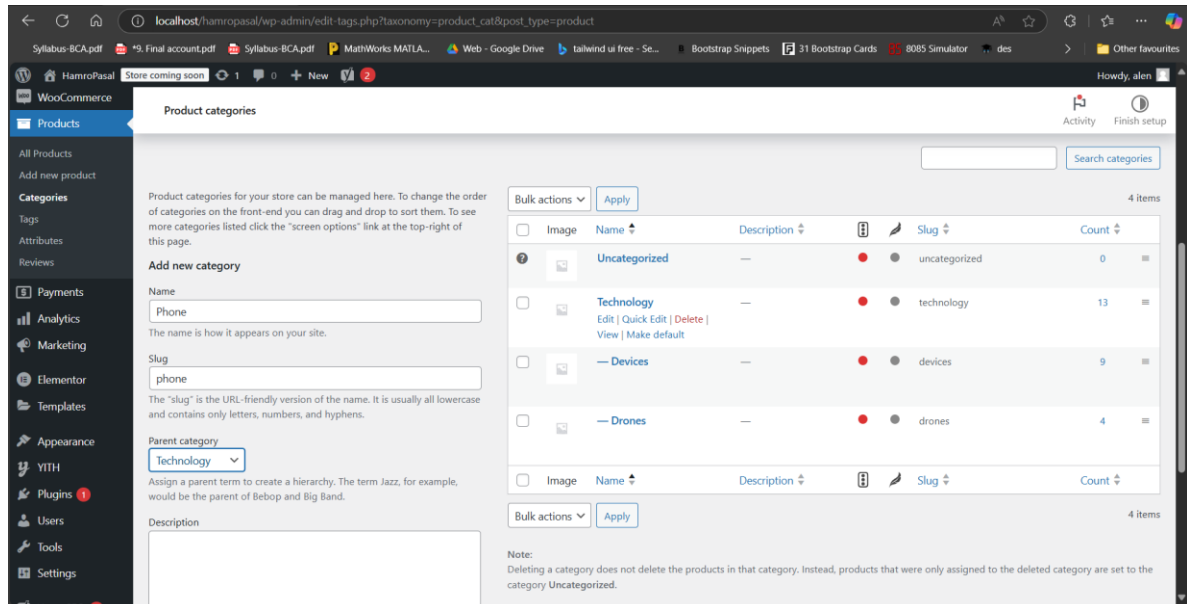


Figure 2.9: Adding Categories

To add a category in WooCommerce on WordPress, the user should go to **Products > Categories** in the WordPress dashboard. Here, they can enter a name for the category, a slug (URL-friendly version), and a description. The user can also upload a thumbnail image for the category, which will appear on the shop page. If needed, the user can assign the category to a parent category to create a hierarchical structure. After filling out the necessary details, the user clicks **Add new category** to save it. This newly created category can then be assigned to products when adding or editing them.

2.2.3 Adding Payments

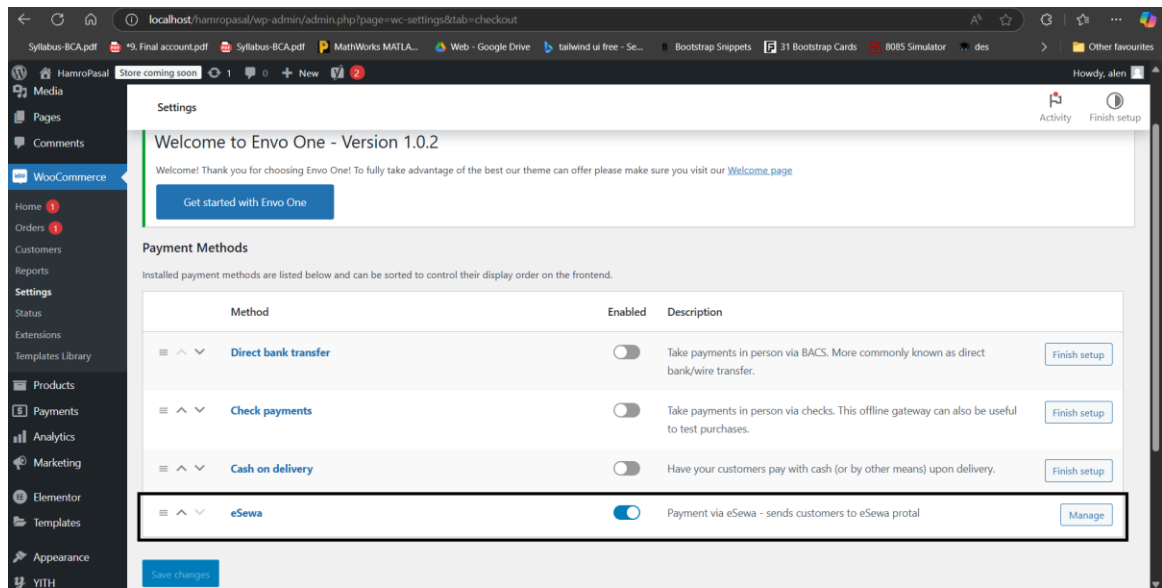


Figure 2.10: Adding Payments

To add a payment option in WooCommerce on WordPress, the user should navigate to **WooCommerce > Settings** and then go to the **Payments** tab. Here, they will see a list of available payment gateways such as eSewa, Cash on delivery, and bank transfers. The user can enable a payment method by toggling the switch next to it, and then clicking the **Set up** or **Manage** button to configure the chosen payment gateway. They will need to enter necessary details, such as API keys for services like ESE eSewa or Stripe, and customize settings like payment instructions and order completion options. After saving the changes, the selected payment option will be available for customers during checkout. [3]

2.2.4 Editing the contents of the Site

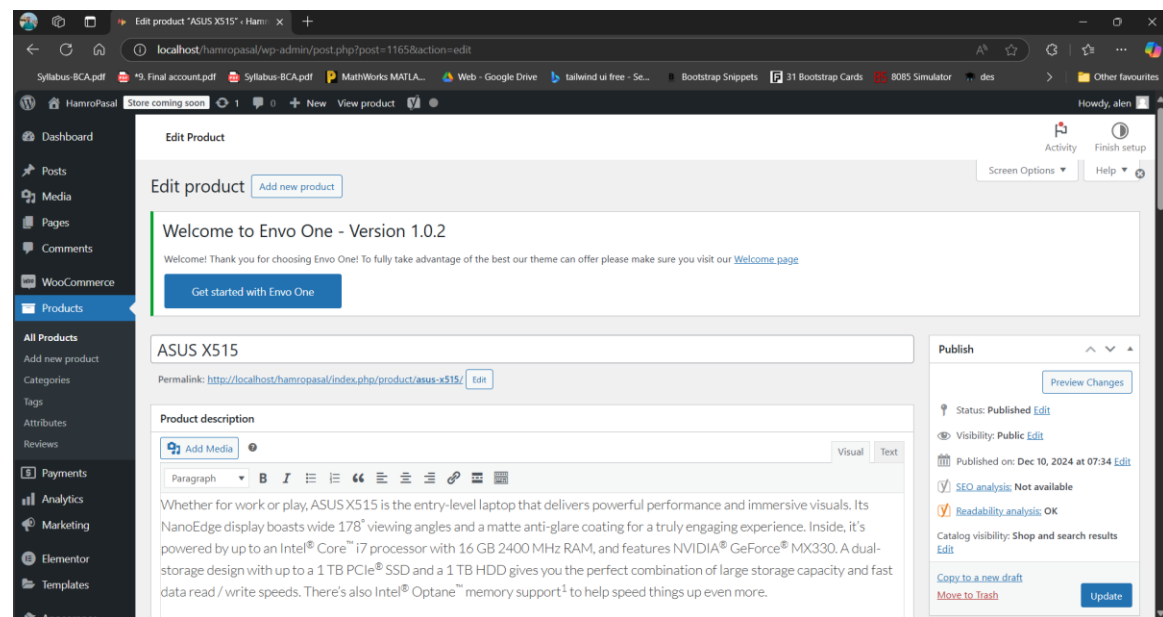


Figure 2.11: Editing the contents of the Site

To edit products in WooCommerce on WordPress, the user should navigate to **Products** > **All Products** in the WordPress dashboard. Here, they can see a list of all products on their site. The user can either click on the product title or the **Edit** link beneath the product to open the product editor. In the editor, the user can update the product title, description, price, SKU, stock status, and images. They can also modify the product's category, tags, and attributes, as well as adjust settings like shipping options and linked products. After making the desired changes, the user clicks **Update** to save the changes and reflect them on the live site.

2.3 Steps of Online Shopping

The following steps are involved in the shopping process:

2.3.1 Browse the shop

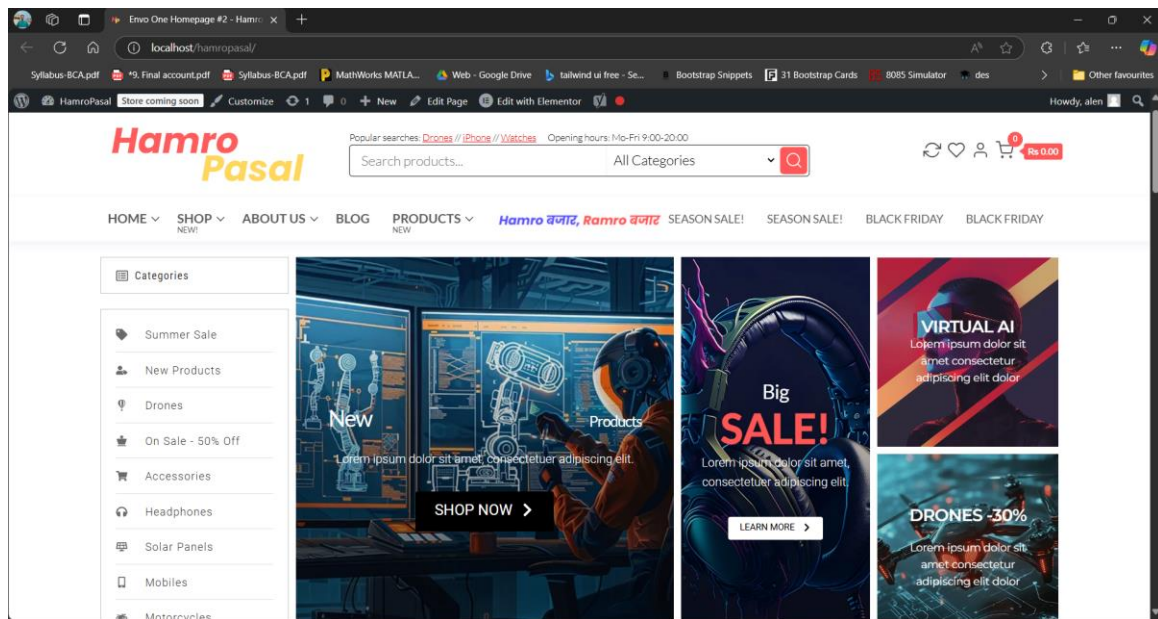


Figure 2.12: Browse the shop

Open the store site and browse the items which can be found using search bar and categories sections. To browse products in a WooCommerce-powered WordPress shop, the user can visit the website's front-end and navigate to the Shop page, which is typically accessible from the main menu or a dedicated link in the site's navigation. Here, they can view all available products, often organized by categories or filters such as price, size, or color. The user can click on any product to see more detailed information, including descriptions, images, pricing, and availability. Additionally, the user may use the search bar or sorting options to quickly find specific products or narrow down the product selection based on preferences.

2.3.2 Add to cart

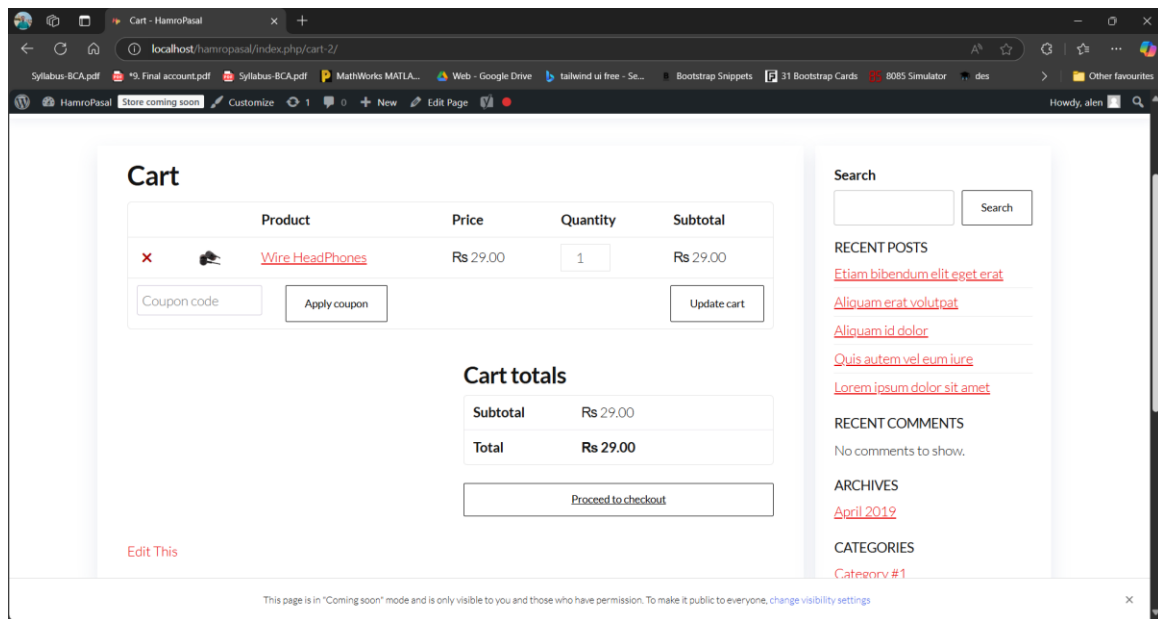


Figure 2.13: Add to cart

In each item there is a option to add item to the cart, using which the item can be added to the cart. To add products to the cart in a WooCommerce-powered WordPress shop, the user can browse the shop page and click on a product they wish to purchase. On the product page, they will select any necessary options, such as size or color, and then click the **Add to Cart** button. Once clicked, the product will be added to the shopping cart, and a confirmation message will appear, showing the product has been added successfully. The user can continue shopping or view the cart by clicking on the cart icon or the **View Cart** button, usually located in the website's header or sidebar.

2.3.3 Checkout and Billing details

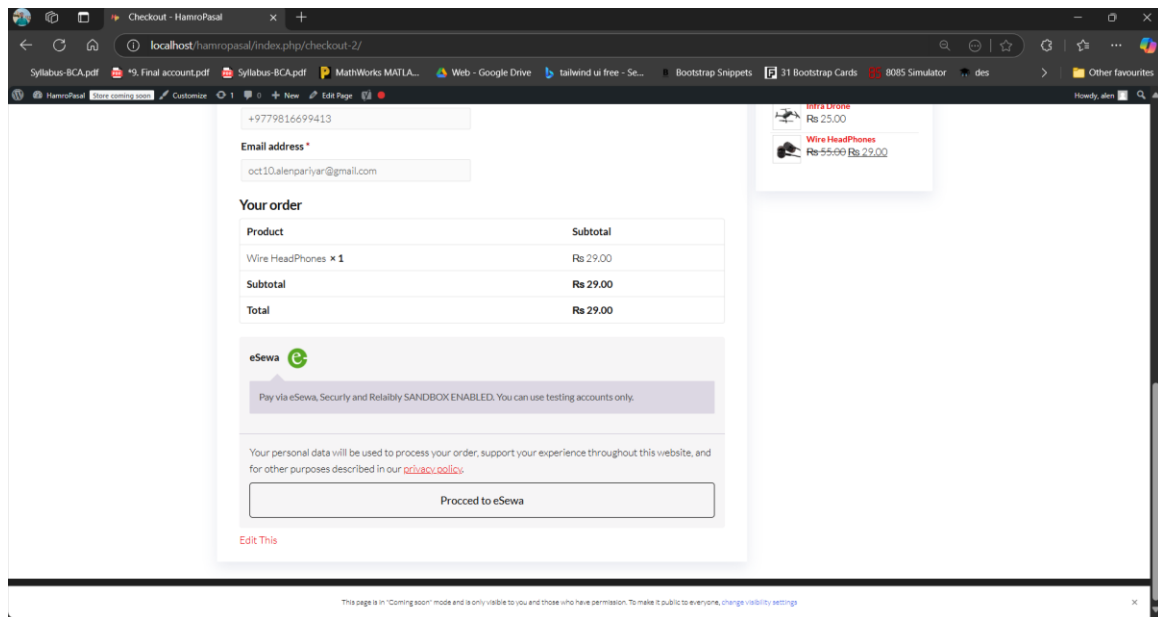


Figure 2.14: Checkout and Billing details

After proceeding with the checkout in a WooCommerce-powered WordPress shop, the user will be directed to the payment page, where they can select from the available payment methods, such as PayPal, credit card, bank transfer, or other enabled gateways. The user simply needs to click on their preferred payment option, enter any required payment details (such as credit card information or Esewa login).

2.3.4 Online Payments

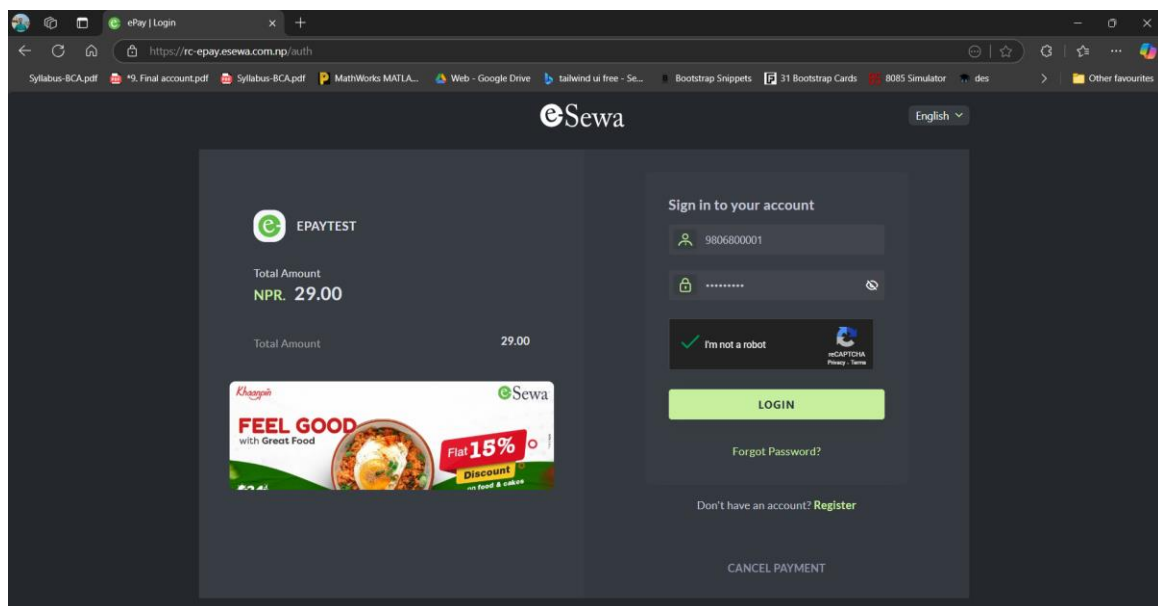


Figure 2.15: Online Payments

After selecting eSewa as the payment method and proceeding to checkout in a WooCommerce-powered WordPress shop, the user will be prompted to enter their **Phone number** and **Password** associated with their eSewa account. Upon entering the correct credentials, they will click **Login** to access their eSewa account. Once logged in, the user can review the payment details and proceed to complete the transaction by following the on-screen instructions. After the payment is successfully processed, the user will receive an order confirmation, and the shop owner will be notified of the completed payment.

2.3.5 Confirm Payment

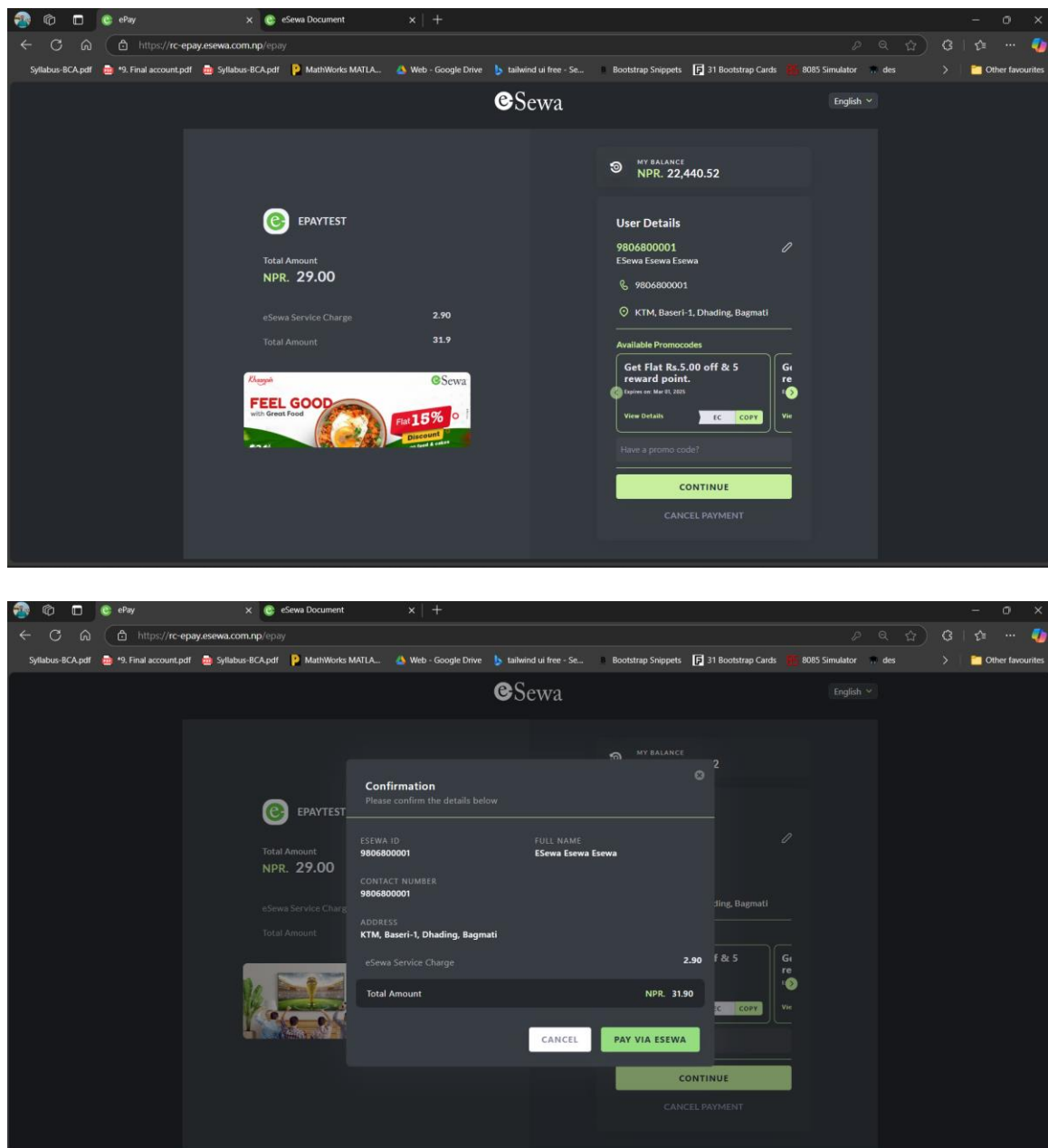


Figure 2.16: Confirm Payment

After logging into the payment gateway system, the user must confirm the payment by reviewing and providing necessary payment information, including their **contact number**, **full name**, **address**, and the **amount** to be paid. The payment gateway will display a summary of the order along with the payment details for verification. The user will need to ensure that all the information is accurate before confirming the payment. Once the details are verified, the user will proceed by clicking the **Confirm Payment** button to finalize the transaction. After successful payment confirmation, the user will receive a notification of the completed transaction, and the shop owner will be informed of the successful payment.

2.3.6 View order info

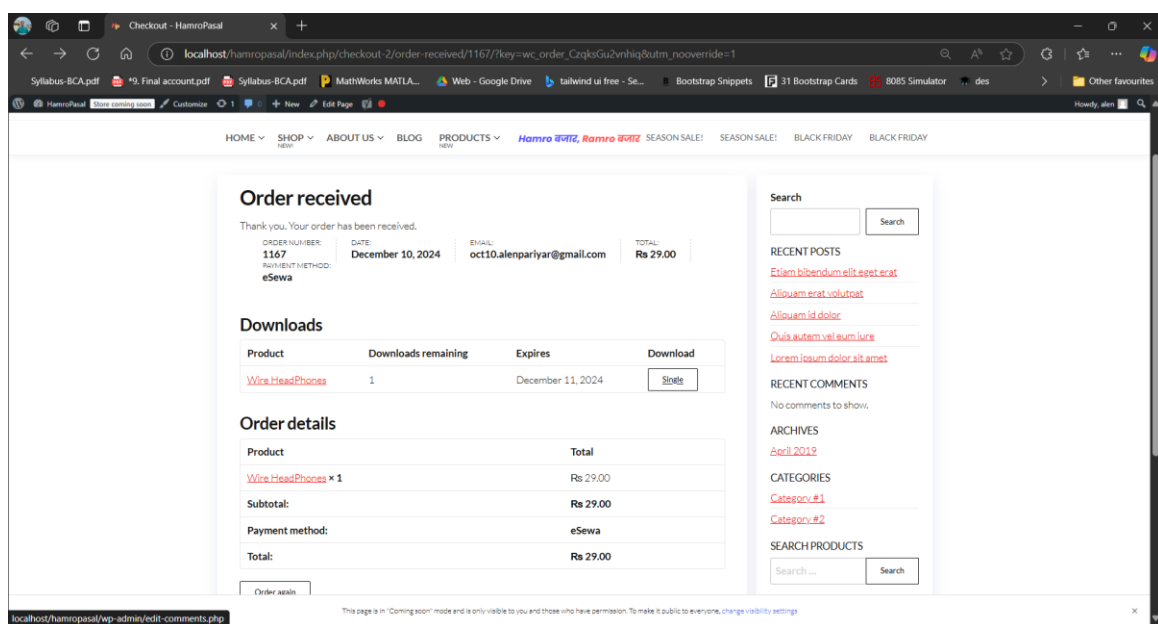


Figure 2.17: View order info

After the payment is completed, the user can view the order status. Here, the user will find details about their purchase. The user can also download the receipt by selecting the specific order, where an option to **Download Receipt** will be available. The receipt will contain important information such as the payment amount, transaction ID, billing details, and the order summary. This allows the user to keep a record of their purchase and track the progress of the order.

2.3.7 Admin view order info

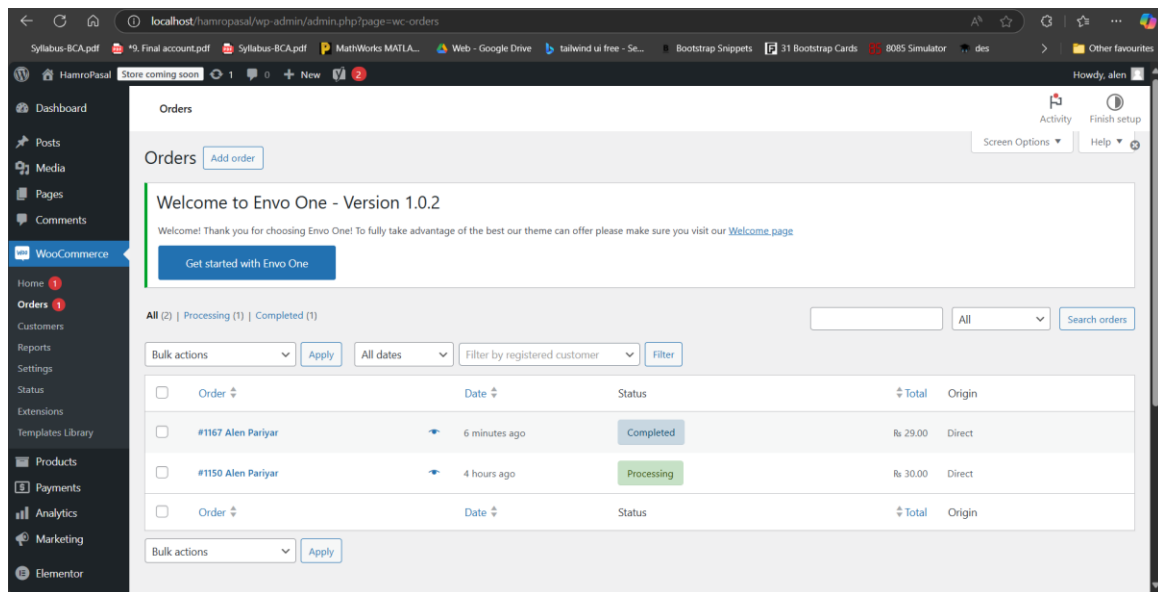


Figure 2.18: Admin view order info

In the admin side of a WooCommerce-powered WordPress shop, the admin can view detailed order information by navigating to the **Orders** section under **WooCommerce**. Here, the admin will see a list of all orders, including key details such as the **order date**, **order status** (e.g., "Processing," "Completed," or "Pending"), the **amount** of the order, and the **method of payment** used (e.g. eSewa, or credit card). The admin can click on any order to view more in-depth information, including the customer's billing details, product items, and shipping information. This allows the admin to manage, update, or fulfill orders efficiently.

Chapter 3: Conclusion and Future Recommendation

3.1 Lesson Learnt

The completion of our group project on developing an e-commerce website using WordPress with eSewa integration has provided us with significant lessons in both technical and collaborative aspects of web development. Firstly, it highlighted the importance of teamwork and effective communication in ensuring all members contribute to achieving a common goal. Secondly, we learned the value of integrating local payment gateways like eSewa to cater to regional user needs while ensuring secure and efficient transactions. Thirdly, the project emphasized the need for a balance between design and functionality to create a user-friendly and visually appealing platform. Additionally, the iterative feedback process from users helped us refine the platform to meet practical requirements. Overall, this project has deepened our understanding of e-commerce development and reinforced the importance of adaptability, collaboration, and innovation in delivering successful digital solutions.

3.2 Conclusion

In conclusion, the successful completion of our group project on an e-commerce website using WordPress with eSewa integration has been a rewarding experience, both technically and personally. This project allowed us to explore the practical applications of MIS and E-Business concepts, bridging theoretical knowledge with real-world implementation. By leveraging WordPress's flexibility and integrating eSewa, we created a platform that meets local market needs while ensuring functionality and user satisfaction. The collaborative effort among team members fostered a deeper appreciation for teamwork and problem-solving. This project has not only enhanced our technical skills but also prepared us for future endeavors in the dynamic field of e-commerce and digital innovation.

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